Md. Moktader Moula

GeoAI & remote sensing researcher · Applying for PhD positions (2027 start)

Chattogram, Bangladesh · ramizmoktader064@gmail.com · ramiz-moktader.github.io/portfolio
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RESEARCH INTERESTS

Geospatial foundation models and satellite embeddings; physics-guided deep learning for surface water mapping; land use and land cover change; spatially honest model evaluation; forest, watershed and climate applications in data-poor regions.

PUBLICATIONS AND MANUSCRIPTS

Islam, A., Pattnaik, N., **Moula, M. M.**, Rötzer, T., Pauleit, S., & Rahman, M. A. (2024). Impact of urban green spaces on air quality: A study of PM_{10} reduction across diverse climates. *Science of The Total Environment*. doi:10.1016/j.scitotenv.2024.176770

Moula, M. M., Shonom, I. J., Islam, A., & Hossain, M. M. (2025). An interactive Google Earth Engine application for global multi-scale vegetation analysis using NDVI thresholding. *arXiv preprint* 2509.01628. doi:10.48550/arXiv.2509.01628

Islam, A., Pattnaik, N., **Moula, M. M.**, Rötzer, T., Pauleit, S., & Rahman, M. A. A remote sensing and statistical modelling framework for assessing the long-term role of neighbourhood urban greenspace to PM_{10} across diverse climate zones. *MethodsX*. UNDER REVIEW

Nandi, R., Jashimuddin, M., Rahman, R., Musa, M. B., **Moula, M. M.**, & Sen Gupta, S. Evaluating the role of Genus *Ficus* in carbon stock and biodiversity in Chattogram, Bangladesh. *Discover Forests*. UNDER REVIEW

Understanding water security in South Asian cities: A study from Nepal and Bangladesh. UNDER REVIEW

Moula, M. M., et al. Comparing PlanetScope, Sentinel-2 and AlphaEarth satellite embeddings for land use and land cover mapping in Chattogram City Corporation. IN PREPARATION · TARGET: SCIENTIFIC REPORTS

Moula, M. M., et al. Physics-guided NIR-absorption attention network for water segmentation in 4-band PlanetScope imagery. IN PREPARATION

Moula, M. M., et al. Advancing surface waterbody mapping in Bangladesh with GeoAI: Fine-tuning the Prithvi geospatial foundation model. IN PREPARATION

Moula, M. M., et al. Impact of land use/land cover transitions on land surface temperature in Chittagong City: A remote sensing and machine learning approach. IN PREPARATION

EDUCATION

Exams 2023 — **MS in Forestry**, Institute of Forestry and Environmental Sciences, University of Chittagong. CGPA 3.56/4.00.
Thesis: An effort to train AI models to identify hilly streams in Sentinel imagery using combinations of normalized difference indices.

Exams 2021 — **BSc (Hons) in Forestry**, University of Chittagong. CGPA 3.47/4.00.
Dissertation: Changes in vegetation cover at the Rohingya refugee camps, 2018–2021.

RESEARCH AND PROFESSIONAL EXPERIENCE

Nov 2024 – now — **Program Officer**, CORE (Cooperation for Resource Efficiency)

- Mapped and analysed more than 500 km² of geospatial data for project planning; built monitoring and evaluation frameworks for 5+ projects; trained and mentored 20 research assistants.

Sep 2023 – Sep 2024 — **Research Assistant**, University of Chittagong · SuLMER; Watershed Management Plans for Teknaf Wildlife Sanctuary and Inani National Park

- Led 15 enumerators surveying 500+ households; analysed 10,000+ records in Python for three watershed management plans.

Oct 2022 – Jan 2024 — **Research Assistant**, University of Dhaka · Impact of the Rohingya influx on wildlife and society

- Analysed survey data from 500+ respondents for a PhD project; prepared three summary reports.

Jul 2022 – Jul 2023	**Research Assistant**, University of Chittagong · Watershed management (Cox's Bazar); Sustainable urban water management in the Hindu Kush Himalaya; USAID ECO LIFE • Produced 10+ maps and visualisations for three projects; contributed analysis to two manuscripts; led field data collection at three sites.
Jul 2021 – Nov 2023	**Research Assistant**, University of Chittagong · Roads and Highways Department PhD project • Processed 100+ Sentinel-2 images in Earth Engine to track five years of land use change; designed a random plot layout for a 300+ plot vegetation survey.
Jan 2022 – Jun 2022	**Intern**, US Forest Service International Programs · Compass project • Made land cover maps for a 400 km² study area; managed a 2,000+ person beneficiary database.
Dec 2021 – Jan 2022	**Research Assistant**, University of Chittagong · Village Common Forests in the Chittagong Hill Tracts • Automated 70+ map layouts with ArcPy.
Aug 2023	**Trainer**, Community Development Centre (CODEC) · three-day data management workshop for 40 staff.

OPEN-SOURCE SOFTWARE

Lead developer	**GeoFlow**, an Earth Engine-style Python framework for Earth observation, spatial statistics and leakage-safe spatial machine learning, with tests and documentation. github.com/ramiz-moktader/geoflow
Developer	**Spatial Point Splitter** (spatially stratified train/test splitting), **GeoPatch AI** QGIS plugin (deep learning patch generation), **KML Grid Generator**.

AWARDS

2022 – 2023	**National Science and Technology (NST) Fellowship**, Ministry of Science and Technology, Bangladesh.

TEACHING AND SERVICE

Dec 2023 – Jun 2024	**Founder and trainer**, free six-month Geospatial Programming Course (JavaScript and Google Earth Engine, in Bengali); 150+ participants from 10+ countries. geocodinglab.com
2013 – now	**Co-founder and board member**, Haate Haat Foundation (education for underprivileged children, disaster relief); taught at Haate Haat School.

SKILLS

Programming	Python (expert), JavaScript, SQL; R and C/C++ (basic)
Machine learning	PyTorch, TensorFlow, TorchGeo, scikit-learn, XGBoost; foundation model fine-tuning
Geospatial	Google Earth Engine (JS & Python), GDAL, Rasterio, GeoPandas, ArcGIS/ArcPy, QGIS, FRAGSTATS
Data	PlanetScope, Sentinel-1/2, Landsat, GEDI, AlphaEarth embeddings; KoboToolbox, Survey Solutions

SELECTED TRAINING

AI, Machine Learning and Deep Learning Research Program (Mahdy Research Academy, 2024); United Nations University Water Learning Center courses on big data for water, active and passive satellite data for flood mapping, spatiotemporal drought assessment, and geospatial analysis with Earth Engine; Introduction to Neural Networks (Simplilearn); Python (Kaggle).

REFERENCES

Available on request, including my MS thesis supervisor at the Institute of Forestry and Environmental Sciences, University of Chittagong.